DYNAMIC ENSEMBLE LEARNING FOR CREDIT CARD FRAUD DETECTION - KDD PROCESS REPORTs

BUSINESS UNDERSTANDING

DATA UNDERSTANDING: *Data Description Report*

- **Location:** The original dataset is available at https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
- **Method used to acquire:** Download the dataset using the `kagglehub` library, which provides a simple way to interact with Kaggle resources.
- **Format:** csv file
- **Quantity:** Rows (284807) - Columns (31)
- **Identities of the fields**: The dataset contains transactions made by credit cards in September 2013 by European cardholders. This dataset presents transactions that occurred in two day. It contains only numerical input variables which are the result of a PCA transformation. Due to confidentiality issues, we cannot provide the original features and more background information. Features V1, V2, … V28 are the principal components obtained with PCA, the only features which have not been transformed with PCA are 'Time' and 'Amount'. Feature 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset.
- **Surfaces features:** Amount, Time, PCs (V1 … V28), Class
  - ***Quantitative Variables***
    - Discrete: -
    - Continuous: Amount, Time, PCs (V1 … V28)
  - ***Categorical Variables***
    - Nominal: Class (0=Legit / 1=Fraud)
    - Ordinal: -

DATA UNDERSTANDING: *Data Quality Report*

- **Missing values Analysis**
  - ***Column-Level Analysis*:** Checked the presence of missing values in each of the columns to understand if some imputation was required. There are 0 columns with missing values.
  - ***Row-Level Analysis:*** Checked the presence of rows with more than 50% of missing values among the columns. There are 0 rows with more than 50% of missing values among the columns.

- **Class Balance:** The dataset presents a very high imbalance, as already know for frauds classification problems: Legit-0 (284315 – 99.827%) / Fraud (492 – 0.173%). In the next steps the data imbalance should be handled by using undersampling or oversampling techniques or by adopting models that can handle this very high imbalance within the dataset.

- **Duplicated Row:** The dataset contains 1081/284807 (0.380%) duplicated rows. In particular, it contains 773 duplicated patterns distributed as follows (611 different patterns repeated 2 times, 66 different patterns repeated 3 times, 81 different patterns repeated 4 times, 10 different patterns repeated 5 times, 1 pattern repeated 6 times, 2 different patterns repeated 9 times, 2 different patterns repeated 18 times). Additionally, we check duplicated both in terms of all features + Class and only looking to the features; in both case we have 1081/284807 (0.380%).
  If we leave duplicates in the dataset we may end up in:
  - Bias in the model - The classifier will "see" repeated evidence of the same row. This gives unfair influence to those examples during training.
  - Risk of overfitting - Especially with a small number of fraud cases, repeating exact Class 1 samples inflates the model's confidence.
  - Invalidation of cross-validation - Duplicates might leak into both training and validation folds. This leads to optimistically biased performance metrics (especially precision/recall).

  Based on these considerations, one solution could be of dropping duplicated rows, keeping each of them just one time.

DATA UNDERSTANDING: *Data Exploration Report*

- **Class Balance:** The dataset presents a very high imbalance: Legit-0 (284315 – 99.827%) / Fraud (492 – 0.173%).

- **Time Analysis:** Time contains the seconds elapsed between each transaction and the first transaction in the dataset
  - The skewness value of approximately -0.036 is very close to zero. This indicates that the distribution is highly symmetrical.
  - The mean (94813.86) is slightly greater than the median (84692.00). This small difference confirms the slight negative skew (mean should be less than the median for negative skew) and indicates that the data has a slightly heavier tail on the positive (right) side than on the negative (left) side, despite the skewness value being negative.
  - The time values range from 0.00 to 172792.00. The middle 50% of the data spans from 54201.50 (25%) to 139320.50 (75%).
  - The standard deviation (47488.14) is relatively large compared to the mean, but the symmetrical nature suggests the spread is consistent with a normal-like distribution, and the min/max values appear reasonable for the overall range.
  - For Legit transactions, we can observe a repetitive wave pattern with two large peaks. This suggests transaction volume follows a diurnal (daily) cycle: possibly office hours and nighttime dips, repeated over two days. This gives us confidence that time reflects real-world temporal behavior.
  - For Fraud transactions, we have much lower volume, but the key insight is that fraud attempts are not uniformly distributed. We can see clusters/spikes in specific time windows. This suggests frauds may be targeted in bursts, likely by bots or coordinated attacks.

- **Transaction Amount Distribution**
  - The distribution of Amount is highly skewed right (approximately 16.98). This means the vast majority of the transactions have very small amounts, and the long, thin tail of the distribution extends far out to the right due to a small number of extremely large transactions. These large transactions are acting as significant outliers.
  - The mean (88.47) is far greater than the median (22.00). The mean is being pulled significantly upwards by the extremely large values in the right tail, while the median remains a robust measure of the typical transaction amount.
  - The standard deviation (250.4) is extremely large compared to the mean. This indicates a long right tail, which is a common indicator of a highly skewed distribution and the presence of extreme outliers.
  - Most transactions are small (75% of transactions are ≤ 77.51).
  - There are zero-amount transactions. This may represent failed/incomplete attempts, errors, or may indicate fraud tests (e.g., verifying a card).
  - Logarithmic Transformation [log(x + 1)] is applied, end up into a more symmetric and usable form. This transformation has successfully normalized the Amount feature, pulling the extreme right tail back towards the center of the distribution. Log transform stabilizes variance, compresses outliers, and makes fraud/legit distributions visually and statistically manageable.

- **PCA Feature Analysis**
  - We don’t have information on how the PCA was computed. Thus, we perform preliminary checks knowing how the PCA works and what is expected:
    - PCA must be performed on centered data (mean=0).
      - All the PCs (V1 … V28) have mean ~ 0, suggesting that data have been centered before applying the PCA (also the kde plots show that the distribution are centered around 0);
    - In PCA we have a variance reduction from the first to the last principal direction.
      - We compute the standard deviation and we can confirm that there is a variance reduction, as expected.
    - Median and quartile are symmetric distributed and centered.

        - Correlation matrix shows that all the PCs have 0 correlation, as expected, since PCA applies orthogonal projections.
    - We use the cumulative explained variance to find a cut-off and find the number of PCs to retain at least 85% (V1 … V17), 90% (V1 … V20) and 95% (V1 … V23) of the total variance.
    - Comparing the kde for each PCs between the two classes emerged:
        - V4, V10, V11, V12, V14, V17 are highly promising candidates for model discrimination and maybe feature selection.
        - V2, V3, V5, V6, V8, V13, V16, V18, V22 may not separate well alone but could help in interaction with others.
        - V1, V7, V9, V19, V20, V21, V23, V28, V15, V26 are probably noise or redundant for classification.

- **Features Relevance to Target class**
    - ***Pearson Correlation***
        - PCA features V11–V17 carry strong linear separability.
        - V11, V12, V14, V17, V10 showed strongest (negative) correlation with fraud class.
        - LogAmount, Hour, Time showed weak or no linear correlation.
        - Amount close to 0 correlation.

    - ***Mutual Information***
        - No feature has extremely high MI (> 0.01), which is expected due to high class imbalance and feature obfuscation (PCA).
        - V17, V14, V12, V11, V10, V27 showed highest MI — top fraud indicators.
        - LogAmount, Hour — Moderate MI (more relevant than many PCA features).
        - Amount (raw) — Less informative than LogAmount.
        - Many PCA features (e.g., V1, V4, V3, V8) — Carry small but non-zero signal.